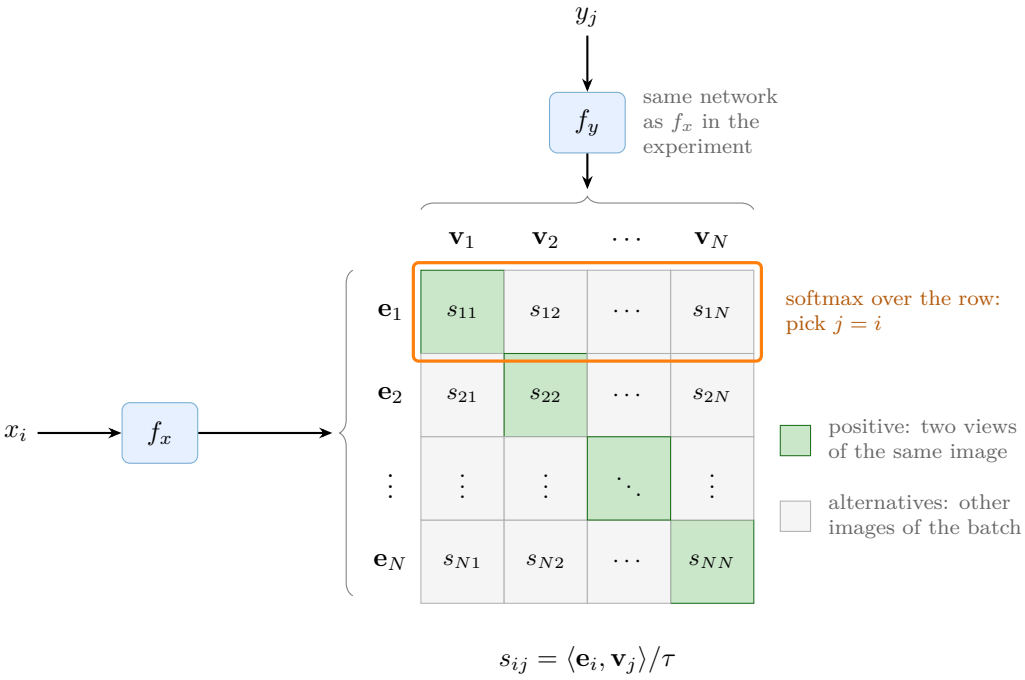

y_j
f_y
same network as f_x in the experiment
$\mathbf{v}_1$ $\mathbf{v}_2$ $\cdots$ $\mathbf{v}_N$
$\mathbf{e}_1$ s_{11} s_{12} $\cdots$ s_{1N}
$\mathbf{e}_2$ s_{21} s_{22} $\cdots$ s_{2N}
$\vdots$ $\vdots$ $\vdots$ $\ddots$ $\vdots$
$\mathbf{e}_N$ s_{N1} s_{N2} $\cdots$ s_{NN}
softmax over the row: pick $j = i$
positive: two views of the same image
alternatives: other images of the batch
x_i
f_x
$s_{ij} = \langle \mathbf{e}_i, \mathbf{v}_j \rangle / \tau$